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**EXPERIENCE**

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**Tunga**

Kampala, Uganda

*Backend Software Engineer (Django)**Nov 2023 - Present*

- **Architected AI Talent Matching and Outreach Agent (Tunga-agent):** Led end-to-end development of a multi-stage agentic pipeline that autonomously classifies companies, matches developers, and runs multi-variant email outreach with engagement-triggered follow-ups. Uses **RAG** with Pinecone vector search and OpenAI embeddings — top-K nearest-neighbour matching with cosine-similarity thresholds and relaxed-criteria fallback — and **GPT-4o-mini** for company ICP classification and personalized outreach. Cut LLM costs **~90%** via hybrid OpenAI/local-Ollama inference, domain-keyed classification caching, and hard pre-filters. Exposed a **Model Context Protocol (MCP)** server surfacing live pipeline analytics; instrumented with **Sentry**, health probes, and a full pipeline audit trail. Deployed on **Kubernetes** with **Celery/Redis**.  
**Core Technologies:** Python, Django, OpenAI, Pinecone, Ollama, MCP, Celery, Redis, SendGrid, Kubernetes, Sentry.
- **Built AI-powered B2B Lead Identification System (Lead-ID):** Designed and led a multi-signal lead-scoring engine detecting **40+ buying-intent signal types across 6 intent streams**, with recency-decay weighting, per-signal caps, and multi-stream reinforcement to rank companies by purchase intent. Generalized the platform to power **three independent verticals** (IT staffing, accounting & finance, AI consulting) via DB-stored, per-use-case ICP rules, signal weights, and thresholds tunable without code changes. Uses GPT-4o-mini for job-description classification and "why now" outreach narratives, with regex fallback when the LLM is unavailable. Processes **500+ companies per batch** with score delta tracking across runs.  
**Core Technologies:** Python, Django, OpenAI, Celery, Redis, PostgreSQL, Clay.
- **Developed an AI-powered resume parser:** Implemented a feature that automatically populates developer profiles during onboarding by intelligently parsing resumes. This improved the efficiency of the profile creation workflow and reduced manual entry for developers.
- **Optimized platform performance and billing:** Diagnosed and resolved "N + 1 query" issues across several backend endpoints, leading to significant API performance improvements. Implemented multi-line and multi-developer invoicing, and expanded payment options to USD (previously EUR-only), reducing currency conversion fees for developers.

**Sunbird AI**

Kampala, Uganda

*Software Engineer and Team Lead**May 2020 - Oct 2023*

- **Leadership:** Provided technical leadership and mentorship to a team of developers and interns, ensuring the successful productization of ML models and effective management of MLOps workflows.
- **Model Deployment:** Spearheaded the deployment of language translation models to an Amazon Sagemaker endpoint, facilitating seamless integration through an API for external users. Later on, led the migration from Sagemaker to GCP Vertex AI. These models power the Sunbird API.
- **Language Translation API:** Developed and led the implementation of the Sunbird AI API, offering language technology services including translation, speech recognition, and text-to-speech in Ugandan local languages. Currently, the API is being utilised by Ugandan companies within their applications and was used in the IndabaX language technology hackathon.
- **Language Translation Frontend:** Revamped the front end of the Ugandan languages translation app, transitioning it from vanilla Javascript to React, thus enabling faster feature development and eliminating a whole class of state-related bugs. This app played a pivotal role in the successful launch of Sunbird AI.
- **Noise Monitoring:** Established the end-to-end infrastructure (including back end and front end) for the Environmental Noise Monitoring System in Kampala and Entebbe. Subsequently, orchestrated the migration of core components from a self-hosted setup to AWS IoT, significantly enhancing reliability and robustness. The Mayor of Entebbe relies on this system for monitoring and mitigating noise in the city.

**Andela**

Kampala, Uganda

*Software Engineer**Aug. 2018 - Apr. 2020*

- **Android development:** Built offline-functionality features for the Converge app (an internal meeting room management mobile app). Also improved how we wrote instrumented tests to mimic offline situations using the Test Butler library.
- **Leadership:** Mentored new developers during their apprenticeship period by assigning them tasks, conducting code reviews and providing technical guidance.

- **Makerere University Data Science and Artificial Intelligence Lab** Kampala, Uganda  
*Software Engineer Intern* *Jun. 2017 – Sept. 2017*
  - **Android development:** Developed a photo-annotation feature in the Android White-Fly count app which helped farmers to annotate pictures of white-fly infested cassava leaves.

## EDUCATION

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- **Makerere University** Kampala, Uganda  
*Bachelor of Science in Computer Science* *Aug. 2015 - Apr. 2018*
- **Coursera Courses** online  
*key specializations*
  - Machine Learning Specialization
  - Mathematics for Machine Learning
  - Deep Learning Specialization

## PROJECTS

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- **AcuGemma:** On-Device AI Tutoring App
  - **Developed an on-device AI tutoring application for primary school students:** Integrated the Gemma 3n 2B LLM directly onto an Android device using MediaPipe, which ensures user privacy, rapid response times, and full offline capability. The application features a dynamic prompting system that guides lessons, evaluates student responses, and provides context-aware feedback.
- **Uganda Coffee Traceability Dashboard:** Geospatial Data Visualization & Verification Tool
  - **Developed a dashboard to verify 97,000+ coffee plots against 2020 satellite baselines.:** Integrated 10m-resolution Google Earth Engine data into a GeoDjango and PostGIS backend, featuring a GraphQL API for real-time coordinate verification and a React-Leaflet interface that manages high-density spatial layers through district-level filtering and geometry optimization.

## PROGRAMMING SKILLS

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- **Languages:** Python, Javascript, Kotlin    **Technologies:** Django, React, FastAPI, PostgreSQL, AWS, GCP, Celery/Redis, Pinecone, OpenAI API, MCP, Kubernetes